

Lesson Notes

MODULE 1 | LESSON 4

SYSTEMIC RISK MEETS MACHINE LEARNING

Reading Time	30 minutes
Prior Knowledge	Entropy, Sensitivity Analysis, Network, Bin size, Lag
Keywords	Transfer Entropy

In the last lesson, we wrapped up a review of some of the most popular and important techniques for monitoring systemic risk in various ways. Suspiciously absent was any innovation since the powerful emergence of machine learning. In this lesson, we rectify that with an overview of the ways that machine learning is being applied in this area of risk management. We also do a deep dive into a network analysis that incorporates both sentiment analysis and price data series to quantify the flow of information between these two "soft" and "hard" domains.

1. Liquidity and Information Flow

Given the higher transaction costs associated with less liquid securities, investors will want to be more confident before buying or selling simply because it costs more. If there are no transaction costs, and you believe there is a 51% chance that the Zenith Bank stock price will increase 3 Nigerian Naira (NGN) and a 49% chance the stock price will decrease 3 NGN, the expectation is positive for buying the Zenith Bank stock. However, if there is a 1 NGN fee to buy or sell Zenith Bank stock, then there is 51% chance that you make a 1 NGN profit and a 49% chance that you lose 5 NGN once you consider the 2 NGN in slippage costs (not factoring in any other brokerage fees): You have to pay 1 NGN to buy the stock and another 1 NGN to sell the stock at the end of the week and make your profit and liquidate your risky position. Of course, in this case, the profit expectation for the trade in this case is firmly negative. And the situation would be even worse if we were factoring in commission fees. Given that we cannot control transaction costs, we assume they are fixed, so I would need to be more confident in the stock's 1 NGN price movement upward in order to affirmatively decide to buy and sell.

This is just a simplified version of one explanation for why less liquid securities markets are likely less responsive to changes in market sentiment and therefore less conducive of information flow more generally. In reality, a single stock transaction is much more likely to exhibit slippage than a much more liquid index. For example, the volume of traded shares in Zenith B might be only a small fraction (say, 1/30) of the volume of the Nigerian Stock Exchange index. The higher turnover volume usually supports much more liquid trading, meaning lower slippage costs. To stretch the "liquidity" metaphor only slightly, if you drop a pebble in a pond, you will see the ripples carry the "information" of that "event" in all directions through the "liquid" water; if you drop a pebble in a puddle of very thick mud, you may see only a slight ripple, if any, due to the illiquidity of this medium.

CDS markets are considered more liquid than bond markets. In fact, research suggests that the introduction of CDS has actually had the effect of decreasing the liquidity of the bond market, for multiple possible reasons (Das 26):

- CDS contracts have "built-in financing" in that they are usually traded on margin.
- For this reason among others, CDSs are traded by hedge funds (which prefer this leverage option) compared to unlevered asset managers. Hedge funds generally have shorter investment horizons than their unlevered counterparts. This again adds to the volume (and thus, liquidity) of the CDS contracts compared to the bond markets.
- Before the introduction of CDSs, both hedge funds and unlevered funds had to trade bonds directly in order to have exposure to the credit market, but since CDSs are available, hedge funds have left the "cash" bond market, resulting in a decrease in volume there.

Bear in mind, too, that liquidity can also be self-fulfilling: Investors prefer more liquid instruments and so they dump their bond positions and pile into CDS, thus making bonds even less liquid and CDS even more liquid.

Given the historically increasing liquidity of CDS and simultaneously decreasing liquidity of bonds, in addition to the above comments about liquidity and information flow, we can easily make sense of the result noted by Scaramozzino et al. that the

importance of information flow from CDS markets has been increasing.

2. Transfer Entropy in Python

In the required readings, you will read about transfer entropy and its relationship to the other information quantification concepts that you already know. For example, Scaramozzino et al. define transfer entropy as "a conditional mutual information between the past of a variable and the future of another variable conditioned to the past of this second variable. It measures the information transferred between the two variables or equivalently the reduction in uncertainty uniquely caused by a variable on the another" (2).

We can experiment with transfer entropy, as well as the other theoretic measures we have learned about, using the very straightforward functions from the PyInform library. So first, let's get the preliminaries out of the way:

```
!pip install pyinform

from pyinform import transfer_entropy
```

Now let's define the following two variables as arrays, where the second one is the same as the first but with a single offset, i.e., the i th element of the first variable is equal to the $i+1$ th element of the second:

```
xs = [0,1,1,1,1,0,0,0,0]

ys = [0,0,1,1,1,1,0,0,0]
```

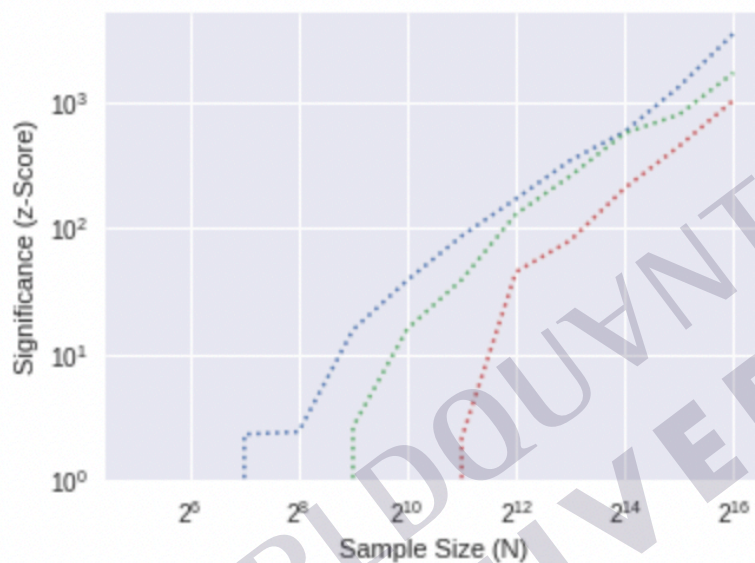
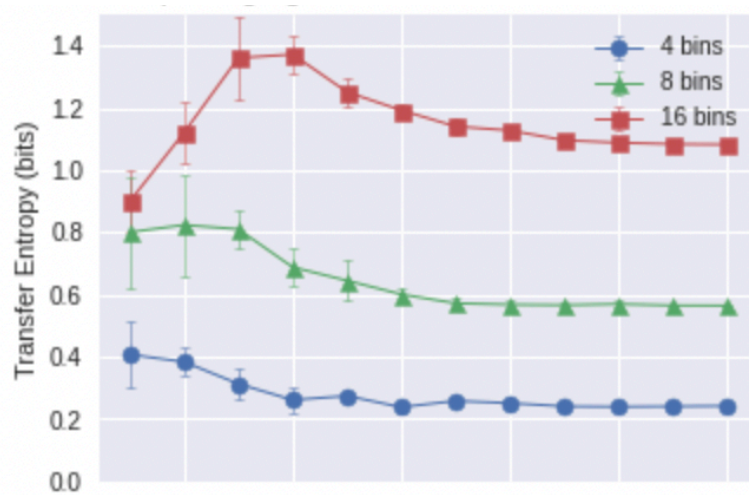
Now we can calculate the transfer entropy between these two with the `transfer_entropy` function. We only need to specify the lag. Since the offset is only one, we should expect that we will obtain the maximum transfer entropy when we set the lag to 1. Let's verify this expectation:

```
transfer_entropy(xs, ys, k=3) yields 0.3333333333333333
transfer_entropy(xs, ys, k=2) yields 0.6792696431662097
transfer_entropy(xs, ys, k=1) yields 0.8112781244591327
```

These results conform to our expectations. Let's go further and explore how bin size affects our estimate of transfer entropy. We note that Scaramozzino et al. observe that "the appropriate choice of bins is crucial" (4). We can use the same Python library that the authors use, PyCausality, to visualize this.

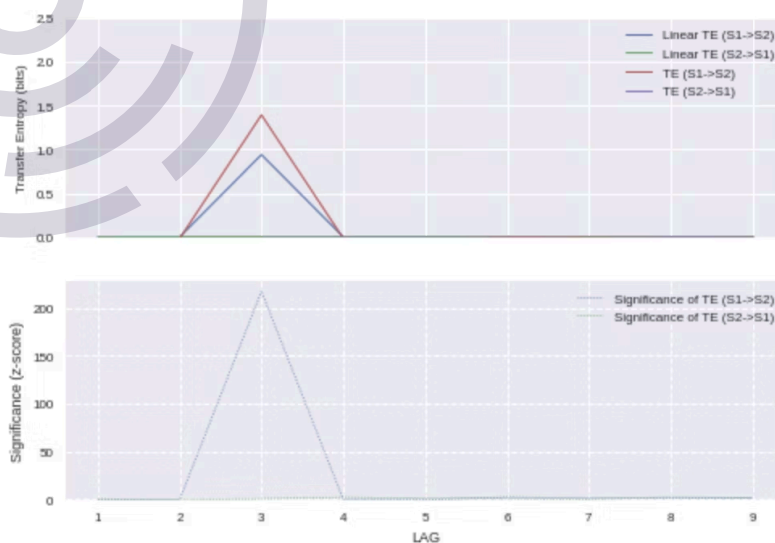
You are encouraged to review the code in the Python script `coupled_map.py` from PyCausality. You will notice that it generates two series, where S_2 is dependent on the S_1 as defined by a quotient α (this is set to 0.4 by default, if you leave the parameter unaltered). When you run this code, the plots make clear that, in this case, the higher number of bins better reveals the Transfer Entropy (top), but that as the sample sizes increase exponentially, the Z-scores for all three of the bin size choices nearly converge (bottom).

Figure 1: Importance of Bin Size in Estimation of Transfer Entropy



We already found in PyInform an easy way to calculate the transfer entropy in Python for a specified lag. And we could easily write some code to loop through this calculation for some range of lags, and then graph these. However, PyCausality provides a nice way to visually inspect the transfer entropy associated with various lags, using both linear and nonlinear versions.

Read through and run the code `te_vs_lag.py`. If you leave the lag between the simulated variable at its default value = 3, then you should see as your output a plot similar to this one:



The plot clearly shows that the lag = 3 has the highest transfer entropy, which is the correct answer since this is the lag we defined for the simulated data. So the code doesn't tell us anything we don't already know in this case, but it helps confirm that we can adapt this code to find the optimal lag between new (non-simulated) data series.

3. Conclusion

In this lesson, we have concluded our survey of systemic risk measures and their Pythonic implementations. In the next module, we tackle a specific systemic risk that is not directly captured by the metrics we have reviewed, nor has it been explicitly mentioned. It nevertheless looms large and provides an example case study for innovative risk management in the face of massive uncertainty.

References

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